

AISSAOUI Ismail

AI & Data Science Engineer

Email: ismail.aissaoui.pro@gmail.com Phone: +213 660 70 77 96

Portfolio: ismail-aissaoui.vercel.app Location: Chlef, Algeria

LinkedIn: linkedin.com/in/aissaoui-ismail-6a77a92a7 GitHub: github.com/i-aissaoui

PROFILE

AI & Data Science Engineer specialized in Deep Learning (Transformers, Mamba) and Edge AI. Proven ability to build end-to-end ML pipelines, optimize model inference, and deploy robust AI solutions that translate technical outcomes into business value.

SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL

ML / DL: PyTorch, TensorFlow, Keras, Scikit-learn, Lightning, ONNX

AI Techniques: Transformers (BERT, GPT, RAG), GNN, GAN, XGBoost, LSTM

Optimization: PSO, Genetic Algorithm, ACO, Simulated Annealing, Bayesian Optimization

Web & API: React, Next.js, FastAPI, Node.js, TailwindCSS

Data & MLOps: MLflow, Airflow, Kubeflow, DVC, Pandas, NumPy

Databases: PostgreSQL, MongoDB, MySQL, Neo4j, Cassandra

Cloud & DevOps: Docker, Linux, Git, CI/CD, CUDA, AWS, Grafana

PROFESSIONAL EXPERIENCE

Graduate AI Research Engineer (Intern)

Jan 2026 – Jul 2026

Sonatrach (Industrial Surveillance Division) — Algeria

Architected an AI-native Digital Twin for real-time monitoring of Gas Turbines.

- Built a Physics-Informed Digital Twin (PINN) using **Mamba-SSM** and **xLSTM** architectures.
- Achieved **0.04ms edge inference** via hardware-software co-design and C++ JIT optimization.
- Engineered custom PINN loss functions to ensure strict thermodynamic consistency.
- Developed quantitative indicators for Out-of-Distribution (OOD) anomaly detection.

Network Engineering Intern

Sep 2024

Algérie Télécom RMC Center — Chlef, Algeria

EDUCATION

National Higher School of Computer Science (ESI-SBA)

2021–2026

Engineering and Master's Degree in Computer Science

Specialization: Artificial Intelligence & Data Science

CERTIFICATIONS

DeepLearning.AI: Machine Learning Specialization, Deep Learning Specialization, NLP Specialization

ZTM Academy: PyTorch: Zero to Mastery, TensorFlow Developer

PROJECTS

Industrial Gas Turbine Digital Twin (Sonatrach)

2026

- **Surrogate Modeling:** Engineered a real-time Physics-Informed Digital Twin using Mamba-SSM and xLSTM.
- **Generative 'What-If':** Implemented generative scenario forecasting for predictive maintenance.
- **XAI & RAG:** Integrated Retrieval-Augmented Generation and Explainable AI for transparent anomaly diagnosis.
- **Edge Optimization:** Achieved 0.04ms edge inference via hardware-software co-design and C++ JIT optimization.

Tech: Mamba-SSM, xLSTM, PyTorch, C++ JIT, PINN, RAG, XAI

Cardiovascular Medical Digital Twin

2026

- **Physics-Informed Modeling:** Architected a Digital Twin modeling 1D Hemodynamic Navier-Stokes flows to predict anomalies.
- **Spatio-Temporal Distillation:** Integrated ST-KDFormer to balance high-fidelity physics with computational efficiency.
- **Data Streaming:** Engineered real-time ingestion of physiological data streams using Apache Kafka.

Tech: PyTorch, PINN, ST-KDFormer, Kafka, Fluid Mechanics

Aura — Enterprise AI Trading Platform

2025

- **Hybrid Forecasting:** Built an autonomous trading system combining LSTM networks for time-series predictions.
- **Strategic Decision-Making:** Implemented Deep Q-Learning (Reinforcement Learning) to optimize long-term reward policies.
- **Real-Time Dashboard:** Engineered a high-throughput FastAPI backend with WebSockets and a Next.js interactive UI.

Tech: TensorFlow, PyTorch, FastAPI, Next.js, WebSockets, RL

GNN-based Recruiting System (Career Connect)

2024–2025

- **Relational Learning:** Developed a recommendation engine using Graph Neural Networks (GNN) to map candidate-job relationships.
- **Semantic Embeddings:** Utilized PyTorch Geometric and SBERT for deep semantic text processing.
- **System Architecture:** Deployed via FastAPI and PostgreSQL, significantly outperforming keyword-based heuristics.

Tech: PyTorch Geometric, SBERT, FastAPI, PostgreSQL, GNN

LANGUAGES

Arabic: Native

|

French: Fluent

|

English: Fluent